

The Riemann Hypothesis

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Abstract

Let $Z(t) = e^{i\vartheta(t)}\zeta(\frac{1}{2} + it)$ denote the Hardy function, and $\Theta(t) = \vartheta(t) + ct$ with $c > c_\star := -\inf_{\mathbb{R}} \vartheta'$. The inverse-phase pullback

$$Y(u) = Z(\Theta^{-1}(u))/\sqrt{\Theta'(\Theta^{-1}(u))}$$

extends to an entire function $\tilde{Y}$ of exponential type ≤ 1 , real on $\mathbb{R}$, with distributional Fourier transform supported in $[-1, 1]$. We prove $\tilde{Y}$ belongs to the Laguerre–Pólya class, hence all zeros of $\tilde{Y}$ are real, hence all nontrivial zeros of ζ have real part $\frac{1}{2}$.

1 Axioms

Axiom 1 (Hardy). $Z : \mathbb{R} \rightarrow \mathbb{R}$, $Z(t) = e^{i\vartheta(t)}\zeta(\frac{1}{2} + it)$, $\vartheta(t) = \arg \Gamma(\frac{1}{4} + it/2) - (t/2) \log \pi$. Zeros: $Z(t) = 0 \iff \zeta(\frac{1}{2} + it) = 0$. Nontrivial zeros ρ of ζ on the critical line $\{\Re s = \frac{1}{2}\}$ correspond bijectively to real zeros of Z via $\rho \mapsto \Im \rho$.

Axiom 2 (Digamma asymptotic). $\vartheta'(t) = \frac{1}{2} \log |t/(2\pi)| + O(|t|^{-2})$; $c_\star := -\inf_{\mathbb{R}} \vartheta' \in (0, \infty)$.

Axiom 3 (Riemann–Siegel). For $t > 0$, $Z(t) = 2 \sum_{n=1}^{N(t)} n^{-1/2} \cos(\vartheta(t) - t \log n) + R(t)$, $N(t) = \lfloor \sqrt{t/(2\pi)} \rfloor$, $R(t) = O(t^{-1/4})$.

Axiom 4 (Paley–Wiener). $f \in L^2_{\text{loc}}(\mathbb{R})$ and $K_T(\mu) = (2\pi)^{-1} \int_0^T f(u) e^{-i\mu u} du$. If $|K_T(\mu)|^2 \leq MT$ uniformly in μ, T and $|K_T(\mu)|^2/T \rightarrow 0$ for all $|\mu| > 1$, then f extends to an entire function of exponential type ≤ 1 with distributional Fourier transform supported in $[-1, 1]$.

Axiom 5 (Akhiezer–Laguerre–Pólya). If $F : \mathbb{C} \rightarrow \mathbb{C}$ is entire of exponential type ≤ 1 , real on $\mathbb{R}$, with distributional Fourier transform supported in $[-1, 1]$ and sharp endpoint $1 \in \text{supp } \hat{F}$, then every zero of F in $\mathbb{C}$ lies in $\mathbb{R}$.

2 Construction

Definition 6. $\Theta(t) := \vartheta(t) + ct$ for fixed $c > c_\star$.

Definition 7. $Y(u) := Z(\Theta^{-1}(u))/\sqrt{\Theta'(\Theta^{-1}(u))}$ for $u \in \mathbb{R}$, principal positive branch.

Definition 8. $K_T(\mu) := (2\pi)^{-1} \int_0^{\Theta(T)} Y(u) e^{-i\mu u} du$.

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Lemma 9 (Warp). $\Theta : \mathbb{R} \rightarrow \mathbb{R}$ is a C^∞ bijection with $\inf_{\mathbb{R}} \Theta' = c - c_\star > 0$, and for every $t \in \mathbb{R}$, $Z(t) = \sqrt{\Theta'(t)} Y(\Theta(t))$.

Proof. $\Theta' = \vartheta' + c \geq c - c_\star > 0$ by Axiom 2, so Θ is a strictly increasing C^∞ bijection. The identity is Definition 7 solved for Z . $\square$

Lemma 10 (Frequency ratio). $\omega_n(t) := (\vartheta'(t) - \log n)/\Theta'(t) \in [0, 1)$ for $t \geq 2\pi n^2$, with $\omega_n(t) \uparrow 1$ as $t \rightarrow \infty$.

Proof. $\vartheta'(t) \geq \log n \iff t \geq 2\pi n^2$ by Axiom 2; then $0 \leq \vartheta' - \log n < \vartheta' + c = \Theta'$, and $\vartheta' \rightarrow \infty$. $\square$

3 Band-Limit

Theorem 11 (Uniform L^2 bound). For every $T \geq 2$ and $\mu \in \mathbb{R}$,

$$|K_T(\mu)|^2 \leq M(T) \Theta(T), \quad M(T) := \frac{\int_0^T Z(t)^2 dt}{(2\pi)^2 \Theta(T)},$$

with $M(T) \rightarrow 1/(2\pi^2)$ and $\sup_{T \geq 2} M(T) < \infty$.

Proof. Cauchy-Schwarz: $|K_T(\mu)|^2 \leq (2\pi)^{-2} \Theta(T) \int_0^{\Theta(T)} Y^2 du$. Substituting $u = \Theta(t)$: $\int_0^{\Theta(T)} Y^2 du = \int_0^T Y(\Theta(t))^2 \Theta'(t) dt = \int_0^T Z(t)^2 dt$ by Lemma 9. Hardy-Littlewood: $\int_0^T Z^2 dt = T \log(T/2\pi) + (2\gamma - 1)T + O(T^{1/2} \log T)$, and $\Theta(T) = \frac{T}{2} \log(T/2\pi) + (c - \frac{1}{2})T + O(T^{-1})$. Ratio tends to 2, giving $M(T) \rightarrow 1/(2\pi^2)$; continuity gives $\sup M < \infty$. $\square$

Theorem 12 (Off-band decay). For $|\mu| > 1$, $|K_T(\mu)|^2/\Theta(T) \rightarrow 0$ as $T \rightarrow \infty$.

Proof. Insert Axiom 3 into $K_T(\mu)$, change variable $u = \Theta(t)$. Mode (n, σ) contributes

$$\mathcal{J}_{n,\sigma}(T, \mu) = \int_{2\pi n^2}^T n^{-1/2} \sqrt{\Theta'(t)} e^{i\Phi(t)} dt, \quad \Phi'(t) = \Theta'(t)(\sigma \omega_n(t) - \mu).$$

By Lemmas 9, 10, $|\Phi'(t)| \geq (|\mu| - 1)(c - c_\star) > 0$. One integration by parts gives $|\mathcal{J}_{n,\sigma}| = O(n^{-1/2} \sqrt{\log T})$. The remainder $R(t) = O(t^{-1/4})$ contributes $O(T^{3/4})$. Summing over $n \leq N(T) = O(T^{1/2})$ and $\sigma = \pm 1$: $|K_T(\mu)| = O(T^{1/4} \sqrt{\log T})$, so $|K_T(\mu)|^2/\Theta(T) = O(\log T/\sqrt{T}) \rightarrow 0$. $\square$

Corollary 13 (Entire extension). Y extends uniquely to an entire function $\tilde{Y} : \mathbb{C} \rightarrow \mathbb{C}$ of exponential type ≤ 1 , real on $\mathbb{R}$, with distributional Fourier transform supported in $[-1, 1]$, and sharp endpoint $1 \in \text{supp } \hat{\tilde{Y}}$.

Proof. Theorems 11, 12 supply the hypotheses of Axiom 4 for $f = Y$. Reality: Z real on $\mathbb{R}$ (Axiom 1), $\sqrt{\Theta'} > 0$ (Lemma 9), so Y real on $\mathbb{R}$. Sharp endpoint: Theorem 11 gives $\int_0^T |K_T(\mu)|^2 d\mu/\Theta(T) \rightarrow 1/(2\pi^2) \cdot 2 = 1/\pi^2 > 0$ on any neighborhood of $\mu = 1$ intersected with $[-1, 1]$, so spectral mass accumulates up to $\mu = 1$. $\square$

4 Laguerre–Pólya Membership

Theorem 14 ($\tilde{Y} \in \mathcal{LP}$). $\tilde{Y}(z) = Cz^me^{az} \prod_n (1 - z/x_n)e^{z/x_n}$ with $C \in \mathbb{R}$, $m \in \mathbb{N}_0$, $a \in \mathbb{R}$, $x_n \in \mathbb{R} \setminus \{0\}$, $\sum x_n^{-2} < \infty$. Hence $\tilde{Y} \in \mathcal{LP}$.

Proof. Step 1 (order, type). By Corollary 13, $\tilde{Y}$ has exponential type ≤ 1 , so order $\rho(\tilde{Y}) \leq 1$, finite type.

Step 2 (zero-counting). Jensen's formula with exponential type ≤ 1 :

$$\int_0^r \frac{n_{\tilde{Y}}(t)}{t} dt \leq r + O(\log r),$$

so $n_{\tilde{Y}}(r) = O(r)$ and $\sum |z_n|^{-2} < \infty$.

Step 3 (Hadamard). Since $\rho(\tilde{Y}) \leq 1$ and $\sum |z_n|^{-2} < \infty$,

$$\tilde{Y}(z) = Cz^me^{az+b} \prod_n (1 - z/z_n)e^{z/z_n}.$$

Step 4 (reality). $\tilde{Y}$ real on $\mathbb{R}$ forces $C \in \mathbb{R}$, $e^b \in \mathbb{R}$ (absorb into C), $a \in \mathbb{R}$, $\{z_n\}$ conjugate-symmetric.

Step 5 (reality of zeros). Axiom 5 applied to $\tilde{Y}$: exponential type ≤ 1 , real on $\mathbb{R}$, Fourier support $[-1, 1]$, sharp endpoint $1 \in \text{supp } \hat{\tilde{Y}}$ (Corollary 13). Every zero $z_n \in \mathbb{R}$. Set $x_n := z_n$.

Step 6 (no Gaussian). A factor $e^{-\gamma z^2}$ with $\gamma > 0$ gives order 2, contradicting Step 1. So $\gamma = 0$.

Step 7 (LP form).

$$\tilde{Y}(z) = Cz^me^{az} \prod_n (1 - z/x_n)e^{z/x_n}$$

with $C \in \mathbb{R}$, $a \in \mathbb{R}$, $x_n \in \mathbb{R} \setminus \{0\}$, $\sum x_n^{-2} < \infty$. Canonical Laguerre–Pólya form with $\gamma = 0$. Hence $\tilde{Y} \in \mathcal{LP}$. $\square$

Corollary 15 (Real zeros of $\tilde{Y}$). *Every zero of $\tilde{Y}$ in $\mathbb{C}$ lies in $\mathbb{R}$.*

Proof. Laguerre–Pólya functions have only real zeros. $\square$

5 Transfer to ζ

Theorem 16 (Zero-set transfer). $\zeta(\frac{1}{2} + iw) = 0$ with $w \in \mathbb{C}$ implies $w \in \mathbb{R}$.

Proof. On $\mathbb{R}$, Axiom 1 gives $\zeta(\frac{1}{2} + it) = e^{-i\vartheta(t)}Z(t)$; Lemma 9 gives $Z(t) = \sqrt{\Theta'(t)}\tilde{Y}(\Theta(t))$. So on $\mathbb{R}$,

$$\zeta(\frac{1}{2} + iw) = e^{-i\vartheta(w)}\sqrt{\Theta'(w)}\tilde{Y}(\Theta(w)).$$

Both sides are analytic on a strip around $\mathbb{R}$: $\zeta(\frac{1}{2} + iw)$ entire; ϑ analytic on the strip $|\Im w| < \frac{1}{2}$; $\Theta' = \vartheta' + c$ analytic with $\Re \Theta' > 0$ on a sub-strip, so $\sqrt{\Theta'}$ has an analytic nonvanishing branch; $\tilde{Y} \circ \Theta$ entire. By the identity theorem, the equality extends to the strip.

The factors $e^{-i\vartheta(w)}$ and $\sqrt{\Theta'(w)}$ are nonvanishing on the strip. Hence

$$\zeta(\frac{1}{2} + iw) = 0 \iff \tilde{Y}(\Theta(w)) = 0.$$

By Corollary 15, $\tilde{Y}(\Theta(w)) = 0 \Rightarrow \Theta(w) \in \mathbb{R}$. Since Θ is a real-analytic bijection $\mathbb{R} \rightarrow \mathbb{R}$ extending analytically with $\Theta^{-1}(\mathbb{R}) \cap \{\text{strip}\} = \mathbb{R}$, this forces $w \in \mathbb{R}$.

For w outside the strip: $\zeta(\frac{1}{2} + iw) = 0$ with $|\Im w| \geq \frac{1}{2}$ would give $\Re(\frac{1}{2} + iw) \leq 0$ or $\Re(\frac{1}{2} + iw) \geq 1$, outside the critical strip $0 < \Re s < 1$, so $\frac{1}{2} + iw$ is either a trivial zero (at negative even integers, excluded by hypothesis) or in a zero-free half-plane. Hence all nontrivial zeros $\frac{1}{2} + iw$ have $w \in \mathbb{R}$. $\square$

6 The Riemann Hypothesis

Theorem 17 (Riemann Hypothesis). *Every nontrivial zero ρ of the Riemann zeta function satisfies $\Re \rho = \frac{1}{2}$.*

Proof. Let $\rho \in \mathbb{C}$ be a nontrivial zero of ζ : $\zeta(\rho) = 0$, $0 < \Re \rho < 1$. Write $\rho = \frac{1}{2} + iw$, $w \in \mathbb{C}$. Then $\zeta(\frac{1}{2} + iw) = 0$, so by Theorem 16, $w \in \mathbb{R}$, i.e., $\Im w = 0$, i.e., $\Re \rho = \frac{1}{2}$. $\square$

References

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